PERSONAL, RELEVANT BACKGROUND AND FUTURE GOALS STATEMENT

When I was in elementary school, my family took annual trips to the Outer Banks, North Carolina, which is where I first fell in love with biology. One year, I found mysterious spherical blue creatures on the beach. In our attempt to identify these organisms, my dad and I scoured the web to no avail. I discovered a new species! Or at least I thought so at the time. When I later learned that these were in fact blue crab larvae, I was not disappointed but fascinated that something so small and inert could metamorphose into something with such a different morphology. This discovery left me in awe of the enigmas of biology and the world around me.

My next step towards a career in science came in high school during a marine ecology class trip to Key Largo, where I had the opportunity to explore a beautiful ecosystem and also witness firsthand the effects of climate change. I saw evidence of coral bleaching as well as declining seagrass meadows, and came to understand how humans can negatively impact ecosystems as well as the dire effects of climate change. I saw before me the interconnected, fragile balance of the world we inhabit, from an entire landscape down to that same blue crab larvae I discovered as a kid. It was then that I resolved to pursue a career in science, so that I may better understand these threads connecting organismal biology with changing environments, driven by climate change.

Since first seeing those blue crab larvae, I have had many experiences that reinforced and shaped my appreciation of multidisciplinary approaches to science. Aside from my ever-growing appreciation for cellular, organismal, and environmental biology, I have also gained a deep appreciation for the resilience with which science must be approached. Indeed, over the years and through my diverse research experiences, my resilience as a scientist has been tested many times. As much as I love the mysteries of our natural world, I have also come to embrace the challenges of untangling experimental setbacks when investigating these mysteries.

INTELLECTUAL MERIT

From molecules to meadows: The inception of my scientific career started in Dr. Michael Timko's lab (Dept. Biology, U Virginia). My duties initially involved exclusively lab maintenance, but my inquisitive nature got the attention of Dr. Timko and soon I was being taught molecular techniques such as how to generate the recombinant tobacco plants that I was initially tasked with maintaining. This led to an independent project, studying the transcriptional regulation of plant trichome development by overexpressing putative transcription factors in recombinant tobacco plants. Through this project, I got the opportunity **to learn and hone many molecular biology techniques such as transgenesis, DNA and RNA extraction and quantitative PCR**. More importantly, these approaches granted me a great appreciation for molecular biology and the tools by which to explore the inner workings of life on earth.

While working in the Timko lab, I also volunteered in the lab of Dr. Karen McGlathery (Dept. Environmental Sciences, U Virginia), helping to restore seagrass meadows in the Chesapeake Bay. Through this experience, I learned the ecological importance of seagrass and its potential to mitigate the effects of climate change due to its ability to sequester carbon. However, due to climate change, seagrass meadows are declining at a rapid rate, hampering their success at carbon sequestration. It was like I was transported back to that fateful day in Key Largo and reminded of the interconnectedness of life on earth.

With my growing fascination for ecological restoration, I was very fortunate to work with Dr. Sandy Wyllie-Echeverria as part of the University of Washington Friday Harbor Laboratories Research Experience for Undergraduates (REU). Under Dr. Wyllie-Echeverria's mentorship, I studied how the sediment-dwelling worm *Abarenicola pacifica* could hamper seagrass restoration by burying seagrass seeds to a depth that would inhibit successful seagrass seedling development. The findings of my studies indicated that the local benthic community could hamper seagrass restoration efforts. My **first-author manuscript** detailing these findings is presently under review and **I have also presented this work at both the Society for Integrative and Comparative Biology** as well as at the **Western Society for Naturalists**.

Integrating molecular biology with aquatic plant and animal health: I was eager to combine the innovative molecular techniques I learned in the Timko lab with my passion for ecological health, gained through my research experiences in the McGlathery and Wyllie-Echeverria groups. To this end, I independently designed a research project that would allow me to combine these two passions. I applied for and received the highly competitive Harrison Undergraduate Research award. Through this fellowship and under the mentorship of Dr. McGlathery, I was able to investigate, at the molecular level, the consequences of a marine heatwave on the germination of two geographically distinct populations of seagrasses. While conducting this work, I devised a novel methodology for extracting RNA from seagrass seeds which had not been previously published. I found that a marine heatwave significantly reduces the viability of seeds, and that different populations of seagrass seeds have differing heat shock protein expression in response to this stress. The presentation of this work received the **Highest Distinction** from the U Virginia Environmental Sciences Distinguished Majors Program and the **Best Undergraduate Talk award** from the Coastal and Estuarine Research Federation conference. A **first-author manuscript** describing these findings is in the final stages of preparation to be submitted to the journal of Aquatic Botany. Successfully realizing this research from start to finish, and receiving such recognition for my work, especially during a global pandemic, has redoubled my passion for research and appreciation for how science can permit the marrying of such different disciplines as molecular biology and field ecology towards impactful research.

My combined research experiences have left me amazed at the resolution with which molecular approaches can reveal the inner workings of biological systems and processes. Thus, before applying to graduate programs, I set out to gain more hands-on experience with fundamental molecular, cellular, and biochemical approaches. I am presently working as a technician in the lab of Dr. Courtney Smith (Dept. Biology, George Washington U), adopting and comprehensively utilizing an array of such approaches to investigate the innate immune systems of the purple sea urchin (*Strongylocentrotus purpuratus*). I have been working on an independent project studying the novel SpTransformer (SpTrf) protein family and their role in the immune response of the purple sea urchin. I discovered that the SpTrf proteins enhance the phagocytosis of foreign targets, and that different SpTrf isoforms recognize a variety of microbes. I am preparing these findings **for a first-authored manuscript** and **presented the data at the North American Comparative Immunology Workshop.** I was also fortunate to participate in another project investigating how sea urchin microbiomes change with disease. I am a contributing author on a manuscript detailing this work, which is **in review by Frontiers in Marine Science**. In addition to a gamut of fundamental molecular techniques for studying gene and protein expression, as well as microscopy techniques for assessing cellular morphology and function, these projects have also served as a platform through which I have been able to master an array of immune assays in a non-model organism.

Global warming and the global amphibian declines: While working in the Smith lab, I learned from our collaborator Dr. Leon Grayfer (Dept. Biology, GWU) about the chytrid fungus *Batrachochytrium dendrobatidis* (*Bd*), which is the leading cause of worldwide amphibian declines and extinctions. Intuitively, I was interested in how climate change will influence the relationship between different frog species and *Bd*. The amphibian skin is in direct contact with these animals' pathogen-rich environments and the Grayfer lab is exploring the efficacies and shortcomings of amphibian skin immune responses to *Bd* infections. With my past experiences and growing research interests in mind, I reasoned that amphibian susceptibility to this emerging pathogen stems at least in part from the physiological tradeoffs imposed by changing climates. After scouring the literature, I learned that indeed one of the prevailing theories states that amphibians experiencing environmental temperatures beyond those to which their immune systems have evolved are compromised in their capacities to deal with *Bd* infections. In talking with Dr. Grayfer he welcomed my suggestion to **explore the molecular mechanisms by which amphibian immune systems become compromised due to changing temperatures** and invited me to join his research group as a PhD student. This graduate project in the Grayfer lab not only represents an important step in my research career but also presents the opportunity to combine the many approaches and ideas that I have explored as a scientist to date. Through this project, I will be able to integrate the molecular techniques and immune assays that I learned in the Timko and Smith labs, with the understanding and appreciation for environmental health that I discovered in the McGlathery and Wyllie-Echeverria labs. Moreover, I stand to learn many new research approaches during my PhD under Dr. Grayfer's mentorship, which I will in turn utilize in my continuing research career goal to improve animal health and offset the ecological consequences of climate change.

BROADER IMPACTS

Past activities: During my REU, Dr. Wyllie-Echeverria encouraged and empowered me to realize that I am a scientist, paving my path towards a career in academia. From him, I learned that scientists are not only researchers, but also mentors. Just as Dr. Wyllie-Echeverria inspired my pursuit of biology, I am committed to helping others build their foundation and appreciation for science. To this end and during my time as an undergraduate student, I **volunteered as a Teaching Assistant for a Developmental Biology course**. This class was not required for Biology majors, and I was thus able to work and mentor students passionate about science. This was a great learning experience and further motivated me towards the kind of academic I hope to one day be. I have had the opportunity to encourage much younger students towards science. Just as that fateful encounter with blue crab larvae changed my life, I hope that my **guest lectures** on sea urchin biology to **elementary school** students help inspire future scientists. This past summer I also **mentored a high school** student from an underrepresented group in the Smith lab, and it was vastly fulfilling to help build her confidence in the lab. I experienced what Dr. Wyllie-Echeverria must have felt when mentoring me, when at the end of the summer the high school student told me that our interactions convinced her to pursue research as a career. I will continue to learn and adopt from my mentors the best approaches to mentoring junior scientists, so that I may help to mentor others during my PhD and one day espouse these approaches with my own students.

Planned activities: As an NSF GRFP fellow I will continue to mentor high school students through the local School Without Walls program and Thomas Jefferson high school internship program, integrating them into the proposed studies, and enriching their appreciation for basic research. I will help Dr. Grayfer train and mentor undergraduate students in the lab through the GWU Undergraduate Harlan Trust Research Program and credit-based undergraduate research, by also including students in the proposed work and inspiring them towards research. Dr. Grayfer and I will start an initiative at GWU geared towards attracting underrepresented groups into research and academia, providing research opportunities for high school and undergraduate students from underrepresented backgrounds, and pairing such individuals with other research labs on campus. To this end, we will reach out to and visit local high schools in the area and advertise this initiative at GWU and other local universities. We will also create a website dedicated to matching prospective underrepresented students with labs that have open undergraduate research positions. Additionally, I will establish a social media presence that draws attention to the growing threat of current and emerging amphibian diseases by way of the website and through social media. Through these outlets, I will share and make public our interdisciplinary efforts and the key research findings from the Grayfer lab (including those of my high school and undergraduate mentees), thus raising public awareness and showcasing the many positive outcomes of NSF-funded research projects for science and STEM education.

My background in molecular biology, ecology, and immunology has given me a unique perspective, and a distinctive skillset. These skills and resources from the NSF GRFP will allow me to study how the changing climate will impact organismal immune response through both ecological and molecular approaches. In addition, the NSF GRFP will allow me to continue my dedication to mentorship, especially to underrepresented groups, and continuing to make the scientific community diverse.